

Chintan Shah

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SUMMARY

Biomedical Engineer specializing in the end-to-end engineering of precision medical devices, with proven industry experience in intraocular lens mechanical analysis, robotic hardware calibration, and verification testing under ISO 13485 regulatory guidelines. Expert at translating clinical specifications into robust, functional product designs and patent-pending technical solutions.

SKILLS AND COMPETENCIES

- **Design & Prototyping:** SolidWorks (CAD), DfX Principles, 3D Printing (FDM), Rapid Prototyping, Mechanism Design, Fixture Design.
- **Quality & Regulatory:** ISO 13485, GxP Compliance, SOPs, Design Controls, FMEA/Risk Management, CAPA, EU MDR, cGMP.
- **Validation & Testing:** V&V Testing, Design Verification, IQ/OQ/PQ Protocols, Test Method Validation (TMV), Bench Testing, Gage R&R.
- **Technical Tools & Data:** Python, Minitab (SPC/Six Sigma DMAIC), GraphPad Prism, InSpec (Micro Vu), Statistical Analysis, Root Cause Analysis.
- **Communication Skills:** Technical Writing & Documentation, Continuous Improvement Facilitation, Problem Solving.

PROFESSIONAL EXPERIENCE

Manufacturing Engineer, *Matic Robots, Sunnyvale, CA*

Mar 2026 – Present

- Lead **Quality Assurance (QA)** and **V&V** processes for robotic hardware and autonomous cleaning systems, making sure that they work reliably and meet safety and durability criteria throughout high-volume manufacturing.
- Perform **functional testing and sensor calibration** for camera-based navigation stacks, identifying, documenting, and resolving hardware software deviations to maintain system integrity.
- Manage the final inspection and shipping procedures for consumer units, using stringent quality control checkpoints to guarantee all goods match set performance criteria before release.

Product Development Engineer, *Yaron Lab, Tempe, AZ*

Jan 2025 – Mar 2026

- Developed a **patent-pending** medical device by leading the end-to-end R&D lifecycle of a novel universal syringe attachment for the TAT2 delivery system. This involved translating user requirements into **SolidWorks assemblies** and manufacturing two functional mechanical prototypes.
- Utilized **FMEA-based risk management and design controls to identify and mitigate mechanical failure modes**, resulting in a >90% increase in spot containment ($p < 0.05$) during ex vivo porcine tissue studies.
- Validated the efficacy of the design across numerous injection conditions by **conducting V&V testing protocols** that demonstrated a statistically significant reduction in volume variance ($p < 0.0001$) compared to conventional syringes.
- Performed **Risk Management (FMEA)** to identify and mitigate mechanical failure modes, ensuring device reliability during ex vivo porcine tissue studies where spot containment improved by over 90% ($p < 0.05$).
- Supported **design verification and generated compliance documentation** matched with **ISO 13485 Design Controls** using GraphPad Prism for sophisticated statistical analysis of experimental trial data

R&D Process Intern, *Alcon, Belmont, CA*

May 2025 – Aug 2025

- Conducted precise **bench testing and Test Method Validation (TMV)** to assess essential mechanical specifications (axial stiffness and haptic height) for Intraocular Lenses (IOL), directly contributing to the design verification of advanced ophthalmic devices.
- Validated measurement system reliability through **Gage R&R** and **Measurement System Analysis (MSA)** using **Minitab**.
- Developed **InSpec (Micro Vu)** scripts to automate quality inspection workflows, transitioning IOL batch testing from sample-based to full 100% inspection of 40-unit batches, resulting in enhanced compliance and increased throughput.
- Analyzed **Process Capability (cP_k)** for high-volume manufacturing lines to identify risks and ensure alignment with **FDA QSR**.
- **Designed and 3D-printed** mechanical model eye to support Train-the-Trainer sessions for laser activation procedures, replicating clinical ocular geometry to serve as a functional training prop.

Research Intern, *Rapid Manufacturing Labs, IIT Bombay, Mumbai, India*

Dec 2019 – Mar 2020

- Generated **mechanical prototypes and 3D models** using SolidWorks and AutoCAD, applying **Finite Element Analysis (FEA)** to minimize structural errors by 20%.
- Fabricated intricate engineering sub-assemblies through **CNC machining and additive manufacturing** to facilitate structural reliability testing.
- Recorded detailed **testing data and engineering documentation to support iterative design** optimization phases.

EDUCATION

Master of Science in Biomedical Engineering | Arizona State University

Bachelor of Technology in Biomedical Engineering | University of Mumbai (Dwarkadas. J. Sanghvi College of Engineering)